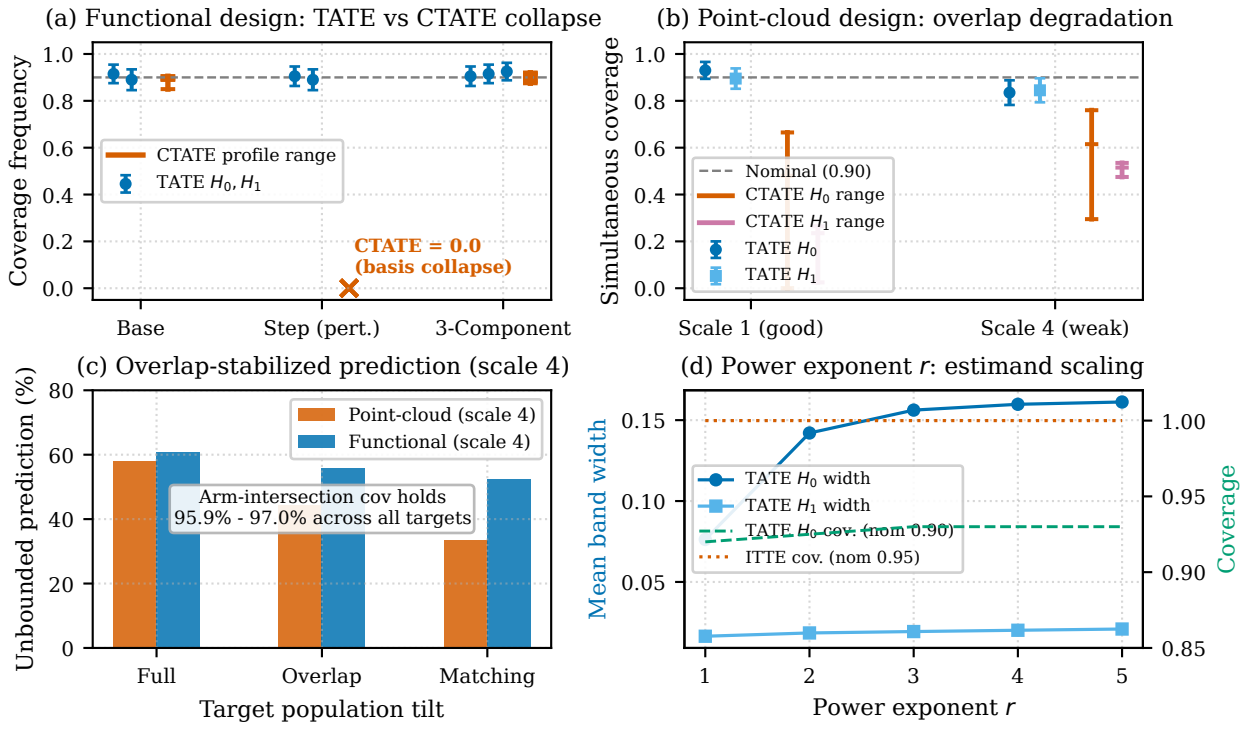
(a) Functional design: TATE vs CTATE collapse
Coverage frequency
CTATE profile range
TATE $H_0, H_1$
CTATE = 0.0
(basis collapse)
Base
Step (pert.)
3-Component
(b) Point-cloud design: overlap degradation
Simultaneous coverage
Nominal (0.90)
CTATE $H_0$ range
CTATE $H_1$ range
TATE $H_0$
TATE $H_1$
Scale 1 (good)
Scale 4 (weak)
(c) Overlap-stabilized prediction (scale 4)
Unbounded prediction (%)
Point-cloud (scale 4)
Functional (scale 4)
Arm-intersection cov holds
95.9% - 97.0% across all targets
Full
Overlap
Matching
Target population tilt
(d) Power exponent $r$: estimand scaling
Mean band width
Coverage
TATE $H_0$ width
TATE $H_1$ width
TATE $H_0$ cov. (nom 0.90)
ITTE cov. (nom 0.95)
Power exponent $r$